GRASSWORKS: Analysis of Bauer et al. (submitted) Functional traits of grasslands: Community weighted mean of specific leaf area (SLA) per plot (esy4)

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Contents

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> **NOTE:** To compare different models, you only have to change the models in the section 'Load models'

Preparation

Protocol of data exploration (Steps 1-8) used from Zuur et al. (2010) Methods Ecol Evol DOI: 10.1111/2041-210X.12577

```r
library(here)
library(tidyverse)
library(ggbeeswarm)
library(patchwork)
library(DHARMa)
library(emmeans)
```

Packages

```r
sites <- read_csv(
  here("data", "processed", "data_processed_sites_esy4.csv"),
  col_names = TRUE, na = c("na", "NA", ""), col_types = cols(
    .default = "?",
    eco.id = "f",
    region = col_factor(levels = c("north", "centre", "south"), ordered = TRUE),
    site.type = col_factor(
      levels = c("positive", "restored", "negative"), ordered = TRUE
      ),
    fertilized = "f",
    obs.year = "f",
    hydrology = "f",
    mngm.type = "f"
  )
) %>%
  mutate(
    esy4 = fct_relevel(esy4, "R", "R22", "R1A"),
    eco.id = factor(eco.id)
    ) %>%
  rename(y = cwm.abu.sla)
```

Load data

Statistics

Data exploration

Means and deviations

```r
Rmisc::CI(sites$y, ci = .95)
```

```
##    upper     mean    lower
## 241.2086 238.5490 235.8893
```

```r
median(sites$y)
```

```
## [1] 238.45
```

```r
sd(sites$y)
```

```
## [1] 33.75003
```

```r
quantile(sites$y, probs = c(0.05, 0.95), na.rm = TRUE)
```

```
##     5%    95%
```

```
## 178.73 293.08
```

```r
sites %>% count(eco.id)
```

```
## # A tibble: 3 x 2
##   eco.id     n
##   <fct>  <int>
## 1 654      203
## 2 664      203
## 3 686      215
```

```r
sites %>% count(site.type)
```

```
## # A tibble: 3 x 2
##   site.type     n
##   <ord>     <int>
## 1 positive    102
## 2 restored    401
## 3 negative    118
```

```r
sites %>% count(esy4)
```

```
## # A tibble: 3 x 2
##   esy4      n
##   <fct> <int>
## 1 R       330
## 2 R22     210
## 3 R1A      81
```

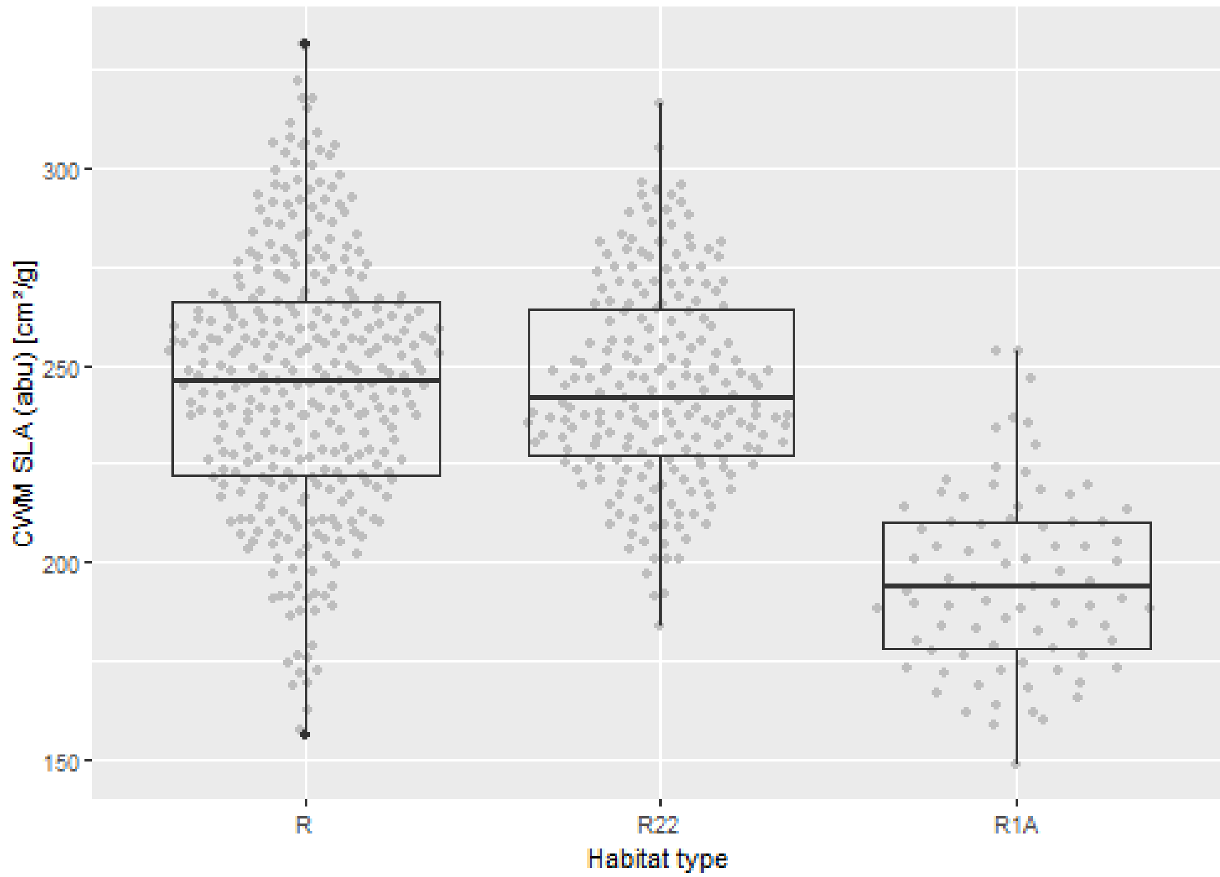
```r
sites %>% count(esy4, eco.id)
```

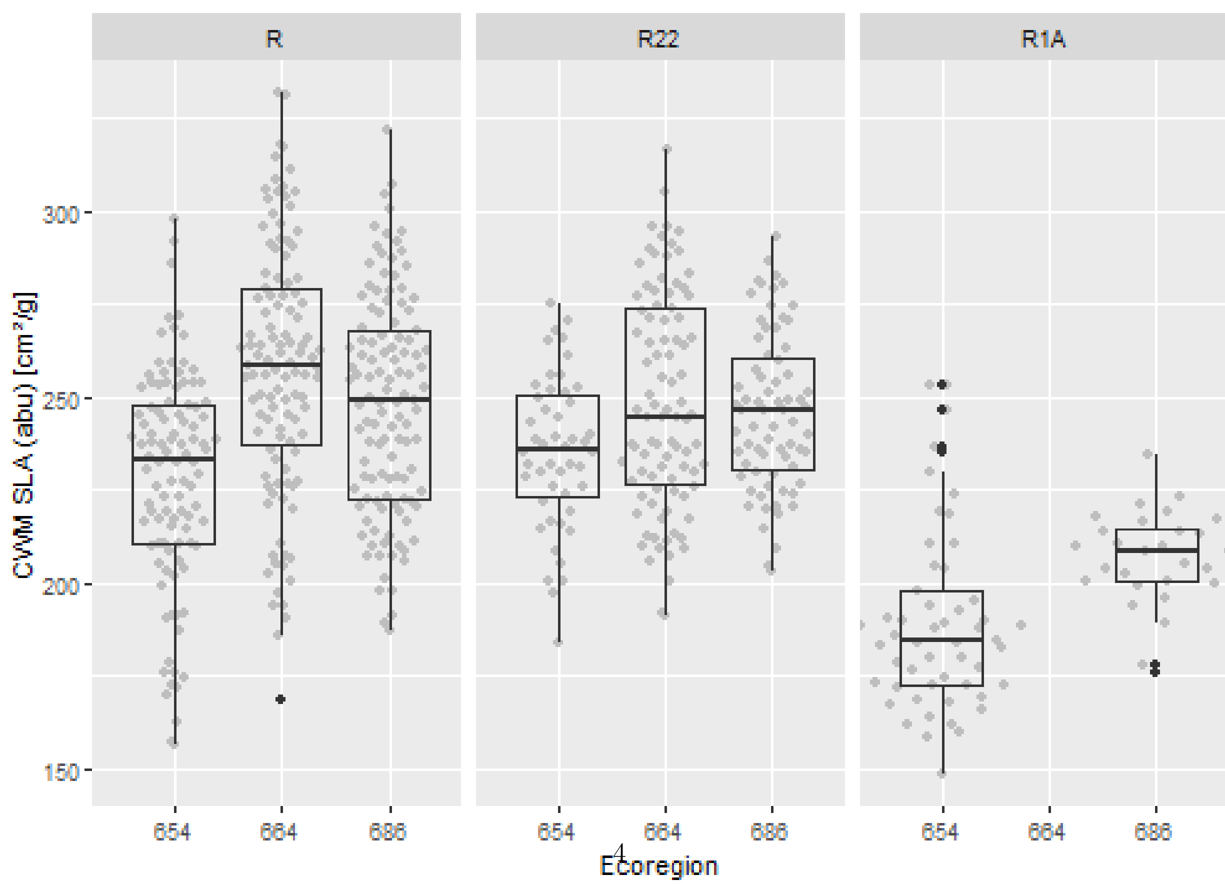
```
## # A tibble: 8 x 3
##   esy4  eco.id     n
##   <fct> <fct>  <int>
## 1 R     654      102
## 2 R     664      112
## 3 R     686      116
## 4 R22   654       48
## 5 R22   664       91
## 6 R22   686       71
## 7 R1A   654       53
## 8 R1A   686       28
```

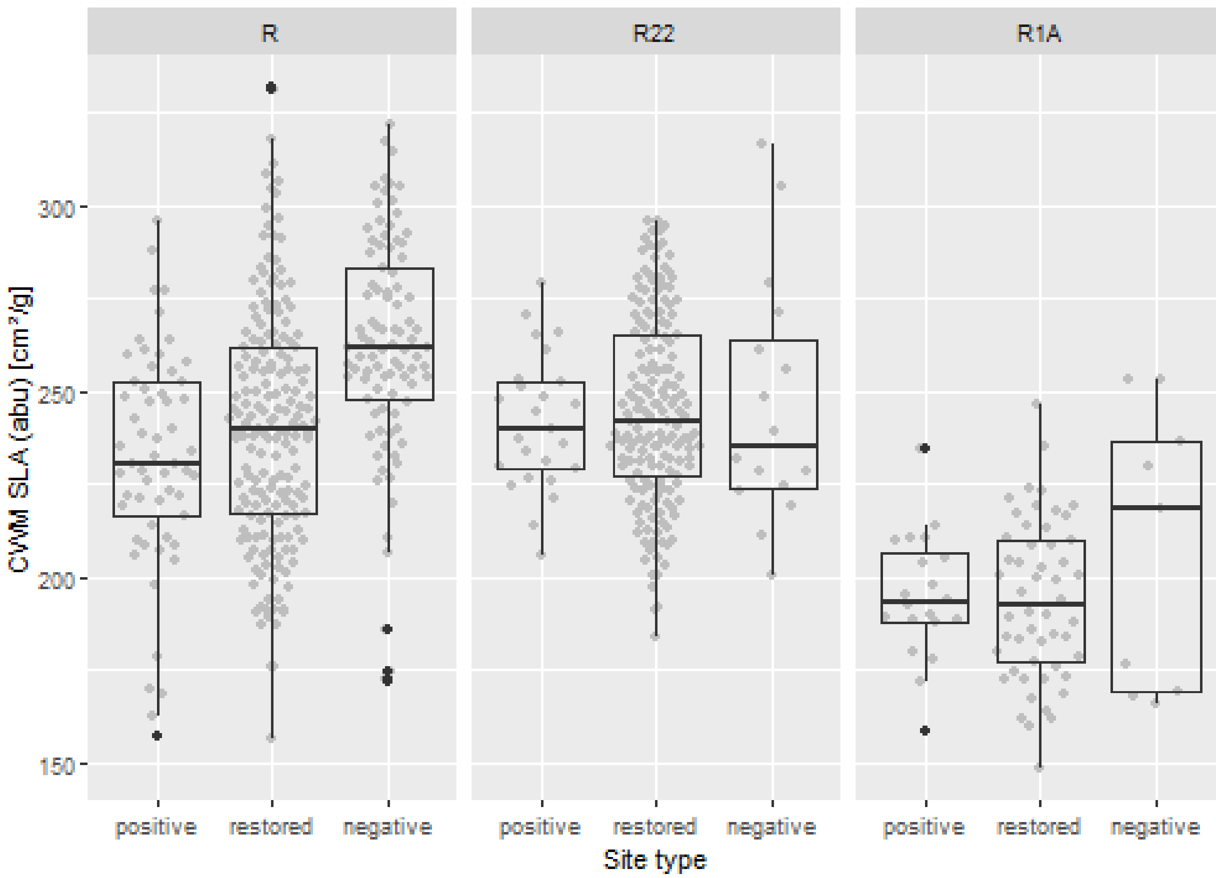
```r
sites %>% count(esy4, site.type)
```

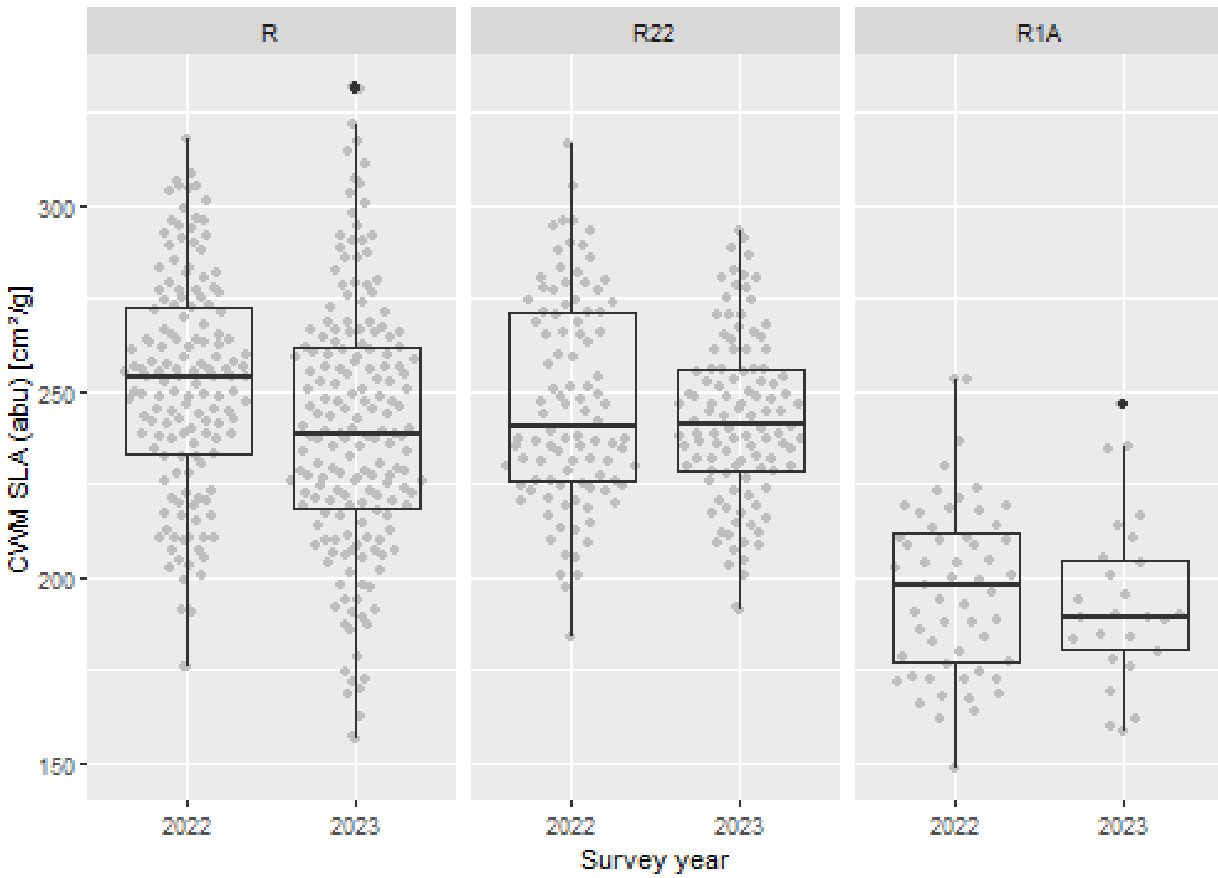
```
## # A tibble: 9 x 3
##   esy4  site.type     n
##   <fct> <ord>     <int>
## 1 R     positive     57
## 2 R     restored    180
## 3 R     negative     93
## 4 R22   positive     25
## 5 R22   restored    169
## 6 R22   negative     16
## 7 R1A   positive     20
## 8 R1A   restored     52
## 9 R1A   negative      9
```

Graphs of raw data (Step 2, 6, 7)



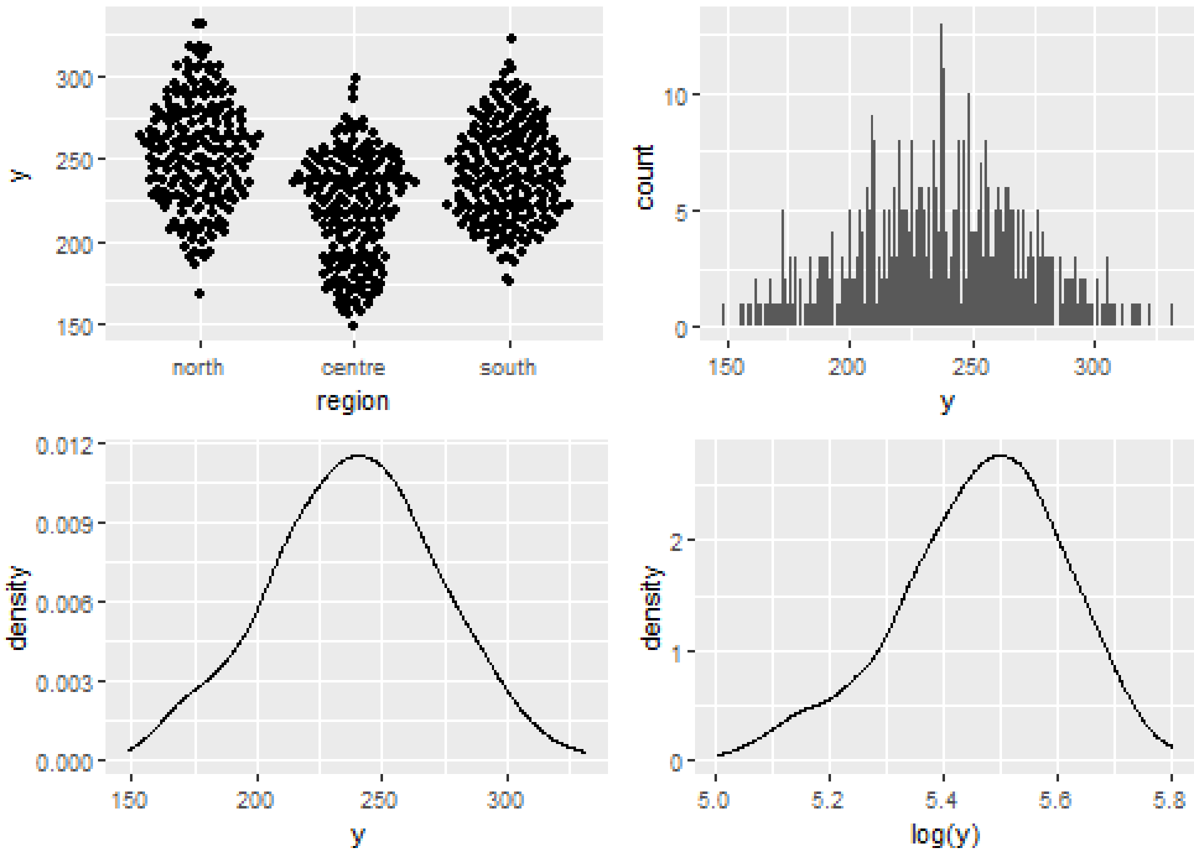



R
R22
R1A
CWM SLA (abu) [cm²/g]
300
250
200
150
positive
restored
negative
positive
restored
negative
positive
restored
negative
Site type



R
R22
R1A
CWM SLA (abu) [cm²/g]
300
250
200
150
2022
2023
2022
2023
2022
2023
Survey year

Outliers, zero-inflation, transformations? (Step 1, 3, 4)



Check collinearity part 1 (Step 5)

Exclude r > 0.7 Dormann et al. 2013 Ecography DOI: 10.1111/j.1600-0587.2012.07348.x

```
# sites %>%
#   select(where(is.numeric), -y, -starts_with("cwm.")) %>%
#   GGally::ggpairs(
#     lower = list(continuous = "smooth_loess")
#     ) +
#   theme(strip.text = element_text(size = 7))

# -> no continuous variables
```

Models

NOTE: Only here you have to modify the script to compare other models

```
load(file = here("outputs", "models", "model_sla_esy4_1.Rdata"))
load(file = here("outputs", "models", "model_sla_esy4_3.Rdata"))
m_1 <- m1
m_2 <- m3
```

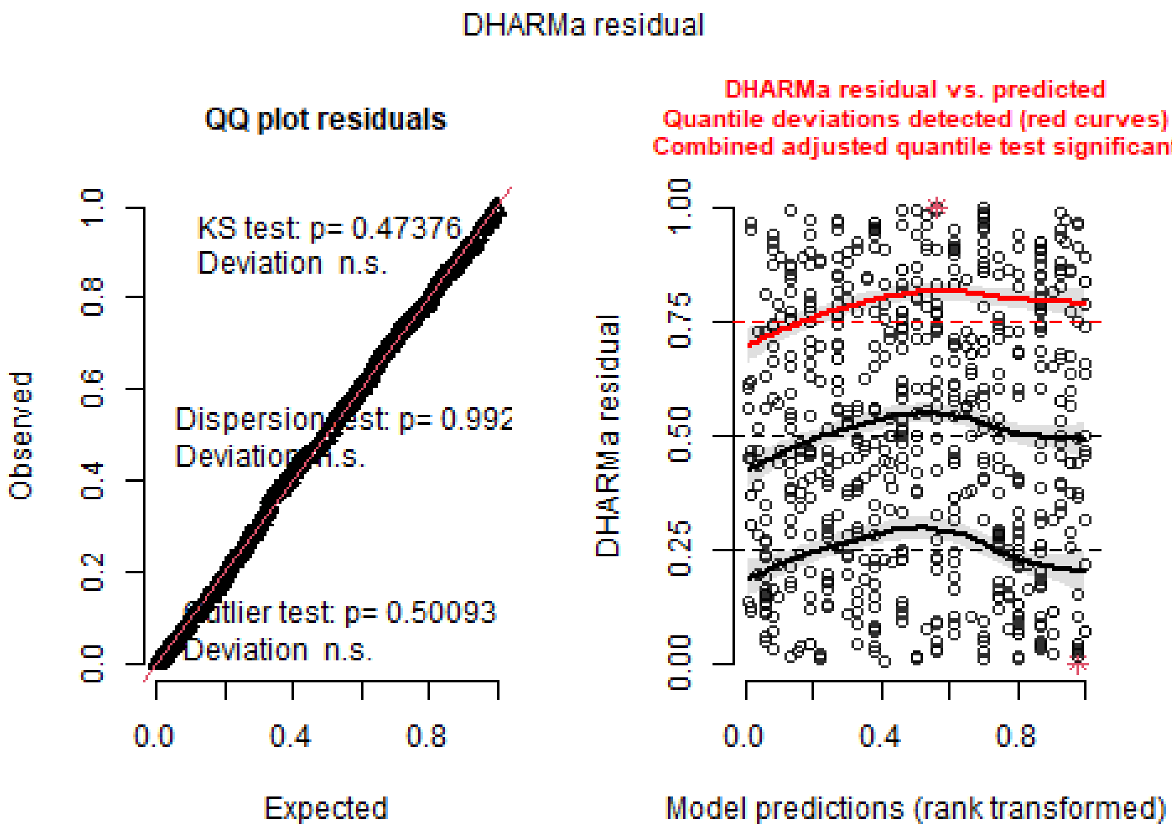
```
m_1@call
## lmer(formula = y ~ esy4 * (site.type + eco.id) + obs.year + (1 |
##     id.site), data = sites, REML = FALSE)
```

```r
m_2@call
## lmer(formula = y ~ esy4 * site.type + eco.id + obs.year + hydrology +
##     (1 | id.site), data = sites, REML = FALSE)
```

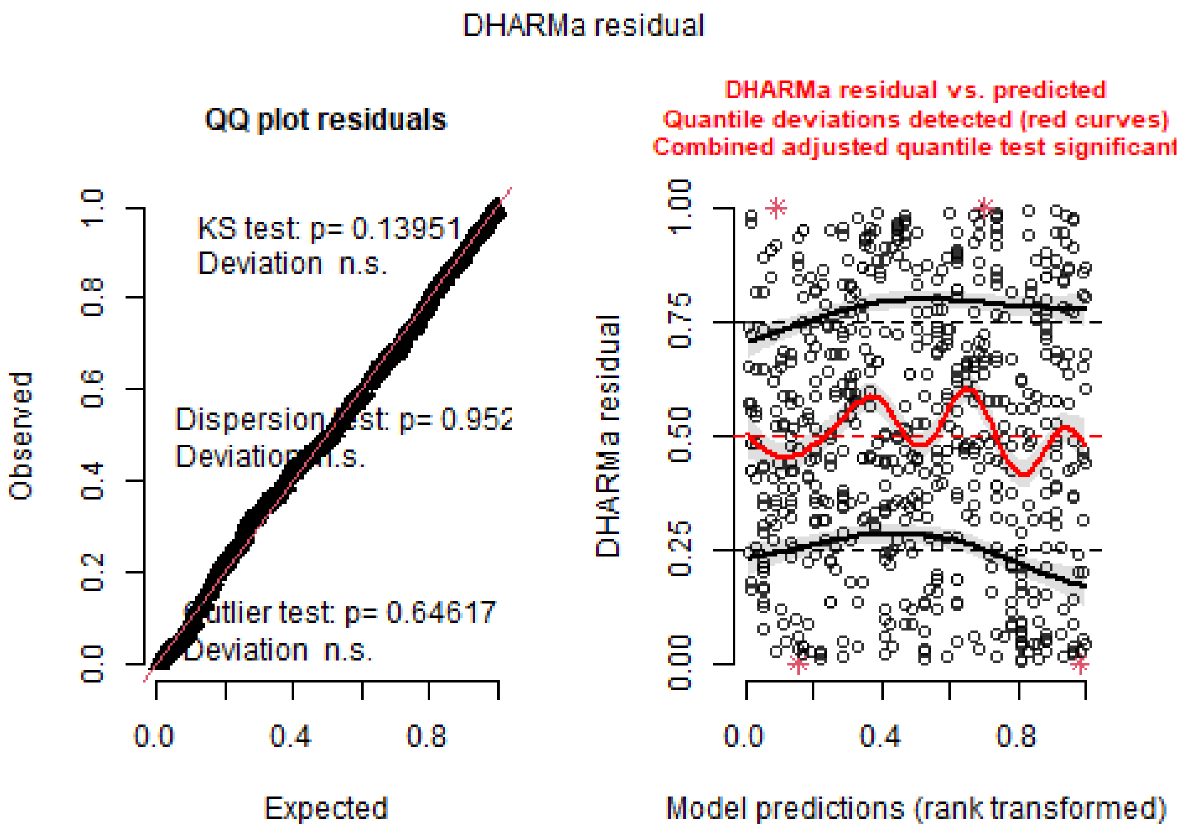
Model check

DHARMa

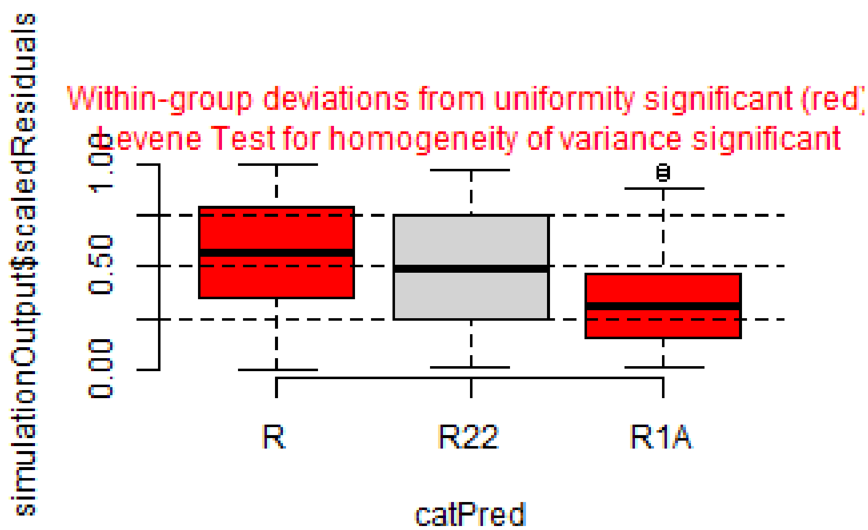
```r
simulation_output_1 <- simulateResiduals(m_1, plot = TRUE)
```



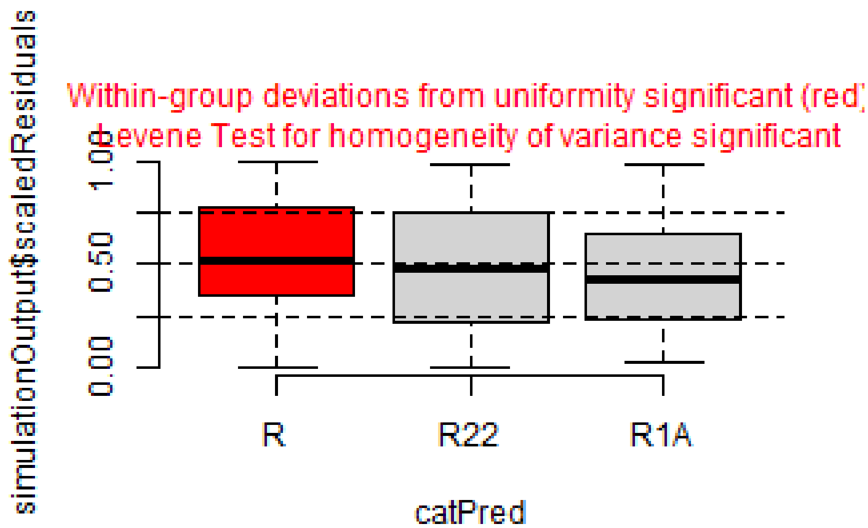

```r
simulation_output_2 <- simulateResiduals(m_2, plot = TRUE)
```



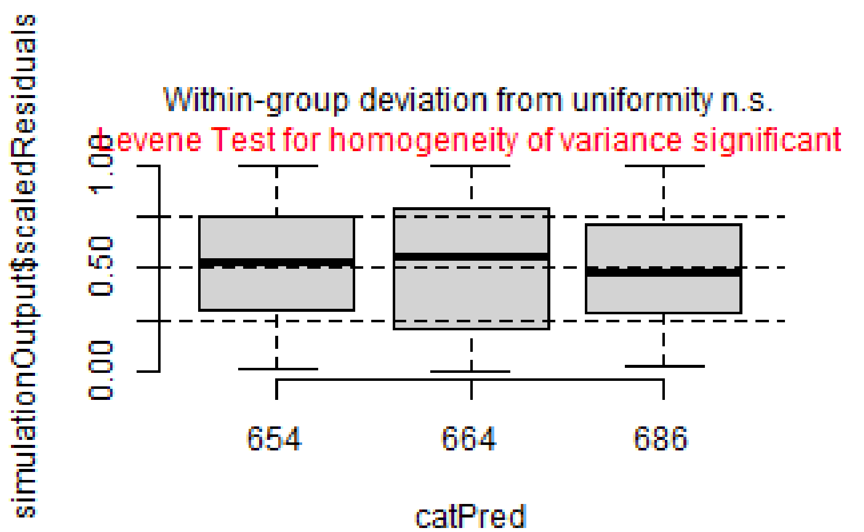

```
plotResiduals(simulation_output_1$scaledResiduals, sites$esy4)
```



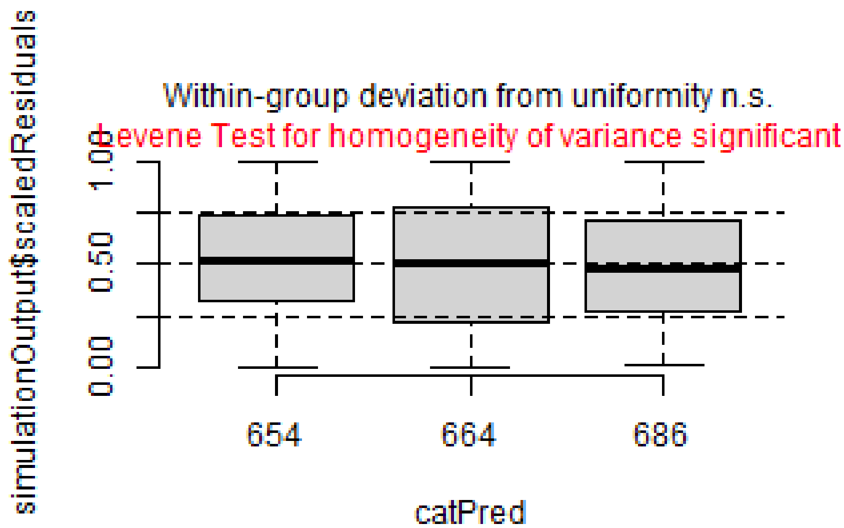

```
plotResiduals(simulation_output_2$scaledResiduals, sites$esy4)
```



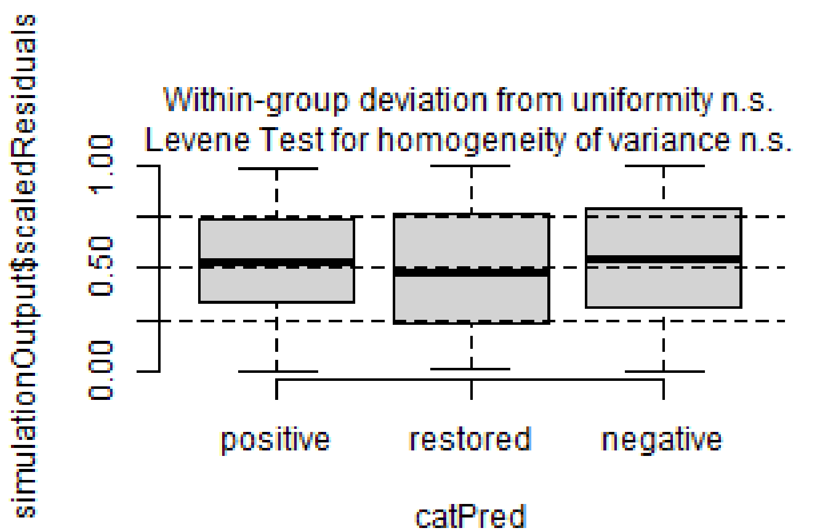

```r
plotResiduals(simulation_output_1$scaledResiduals, sites$eco.id)
```



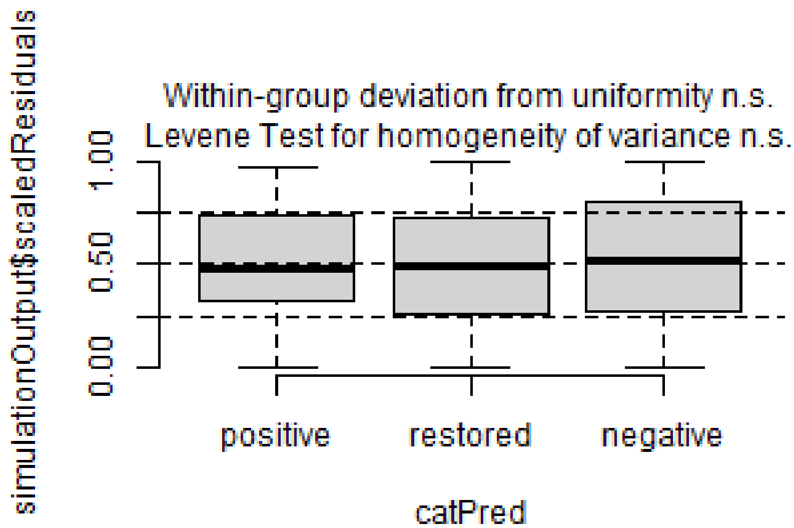

```r
plotResiduals(simulation_output_2$scaledResiduals, sites$eco.id)
```



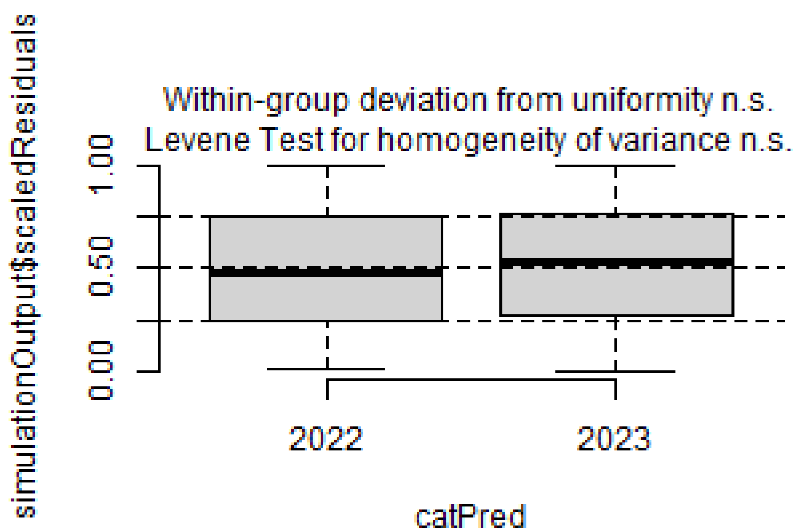

```r
plotResiduals(simulation_output_1$scaledResiduals, sites$site.type)
```



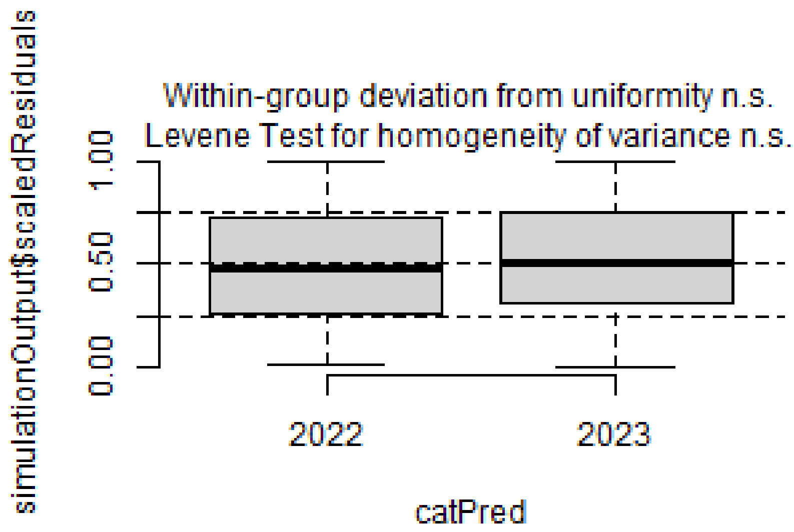

```r
plotResiduals(simulation_output_2$scaledResiduals, sites$site.type)
```



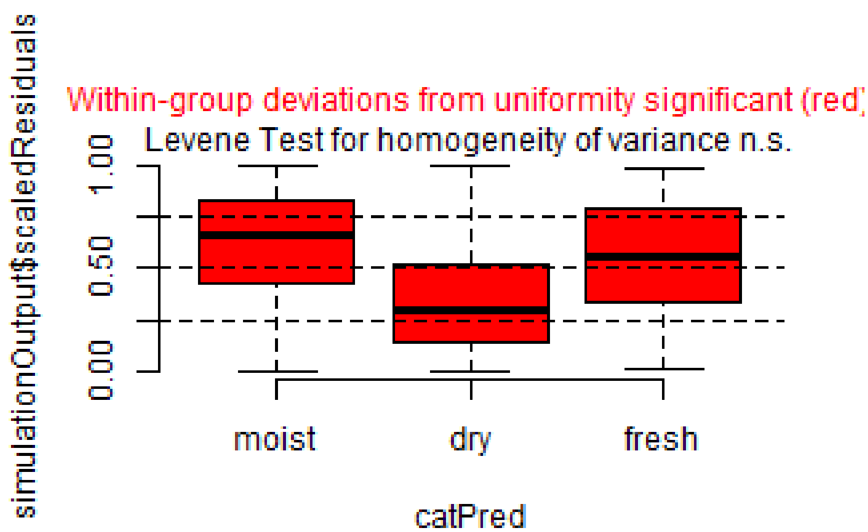

```r
plotResiduals(simulation_output_1$scaledResiduals, sites$obs.year)
```



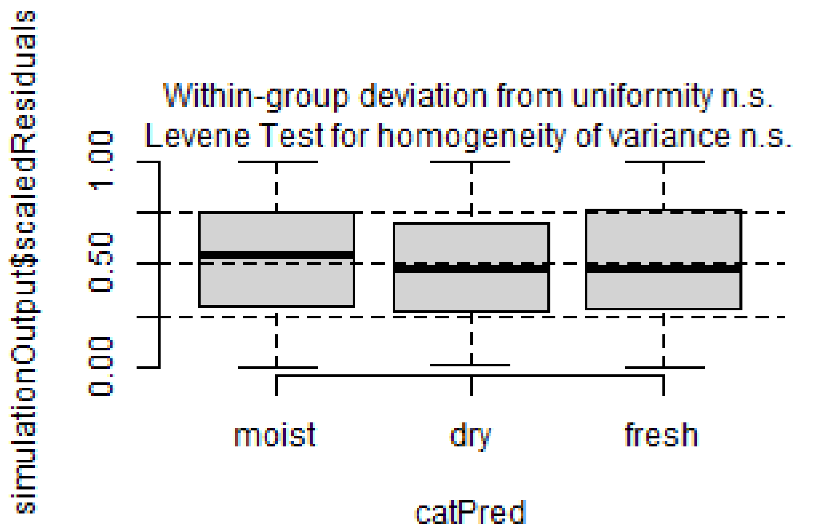

```r
plotResiduals(simulation_output_2$scaledResiduals, sites$obs.year)
```



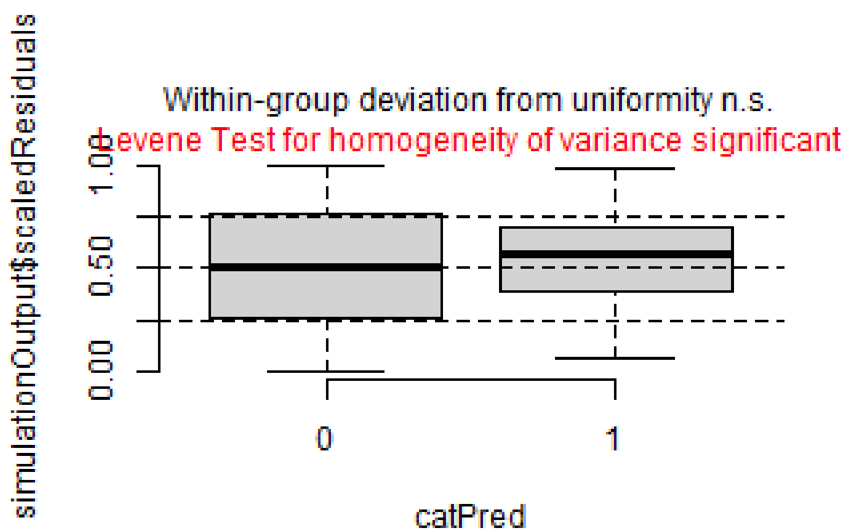

```
plotResiduals(simulation_output_1$scaledResiduals, sites$hydrology)
```



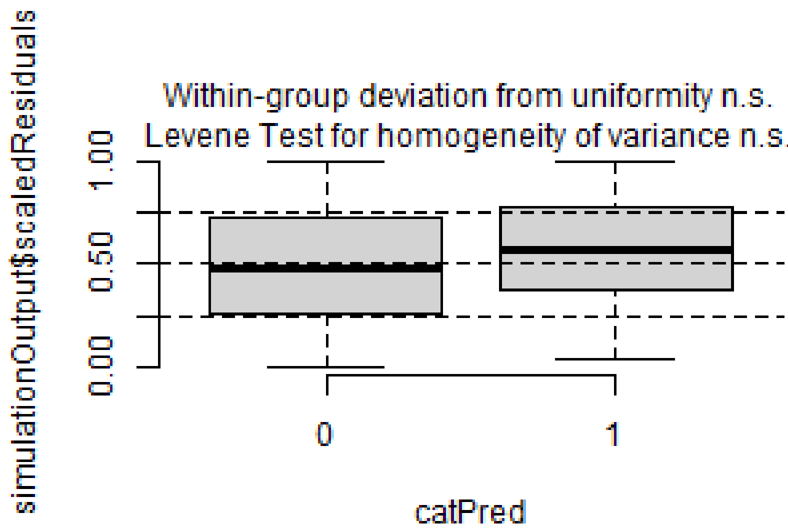

```
plotResiduals(simulation_output_2$scaledResiduals, sites$hydrology)
```



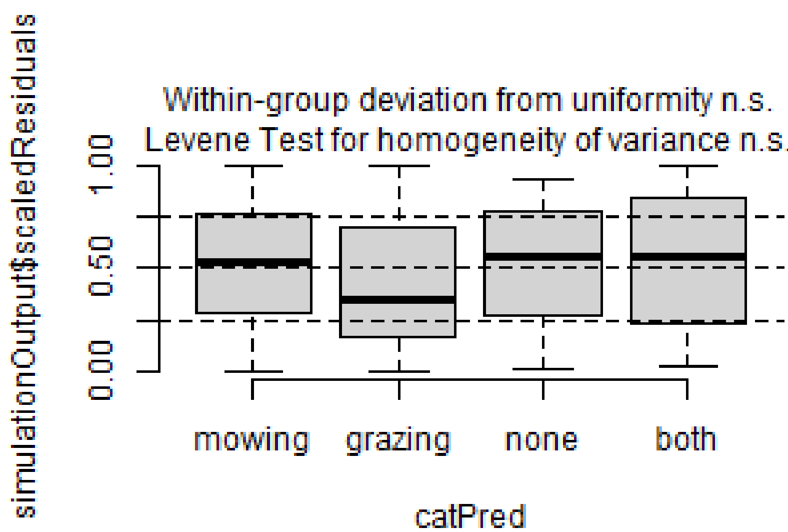

```r
plotResiduals(simulation_output_1$scaledResiduals, sites$fertilized)
```



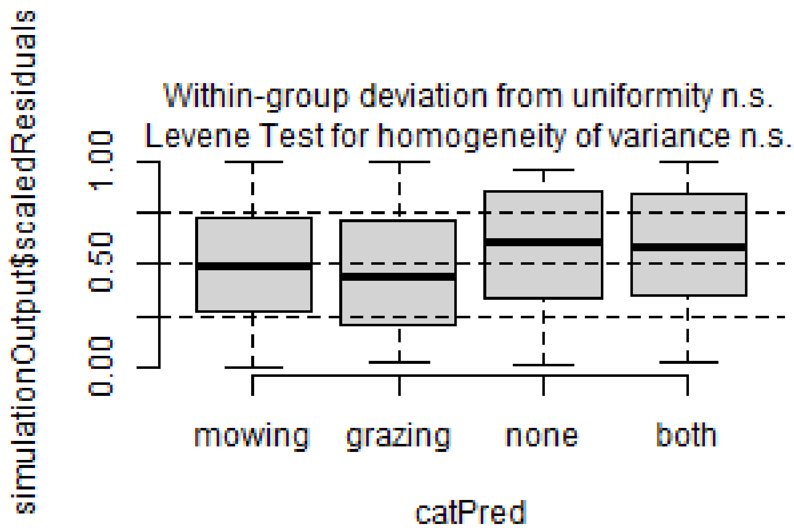

```r
plotResiduals(simulation_output_2$scaledResiduals, sites$fertilized)
```




```r
plotResiduals(simulation_output_1$scaledResiduals, sites$mngm.type)
```




```r
plotResiduals(simulation_output_2$scaledResiduals, sites$mngm.type)
```



Check collinearity part 2 (Step 5)

Remove VIF > 3 or > 10 Zuur et al. 2010 Methods Ecol Evol DOI: 10.1111/j.2041-210X.2009.00001.x

```
car::vif(m_1)
```

```
##                      GVIF Df GVIF^(1/(2*Df))
## esy4            11.414767  2        1.838090
## site.type        1.371237  2        1.082127
## eco.id           1.524788  2        1.111226
## obs.year         1.020636  1        1.010265
## esy4:site.type   4.925531  4        1.220553
## esy4:eco.id      8.915162  3        1.439975
```

```
car::vif(m_2)
```

```
##                     GVIF Df GVIF^(1/(2*Df))
## esy4            3.939149  2        1.408804
## site.type       1.483165  2        1.103564
## eco.id          1.080657  2        1.019581
## obs.year        1.027422  1        1.013618
## hydrology       1.268040  2        1.061166
## esy4:site.type 4.626501  4        1.211035
```

Model comparison

R2 values

```
MuMIn::r.squaredGLMM(m_1)
##            R2m       R2c
## [1,] 0.3313799 0.7767097
MuMIn::r.squaredGLMM(m_2)
##            R2m      R2c
## [1,] 0.4233399 0.790452
```

AICc

Use AICc and not AIC since ratio n/K < 40 Burnahm & Anderson 2002 p. 66 ISBN: 978-0-387-95364-9

```
MuMIn::AICc(m_1, m_2) %>%
  arrange(AICc)
##     df     AICc
## m_2 16 5538.950
## m_1 17 5568.802
```

Predicted values

Summary table

```
car::Anova(m_2, type = 3)
```

```
## Analysis of Deviance Table (Type III Wald chisquare tests)
##
## Response: y
##                    Chisq Df Pr(>Chisq)
## (Intercept)    1903.1430  1  < 2.2e-16 ***
## esy4              8.7355  2    0.01268 *
## site.type        28.3086  2  7.126e-07 ***
## eco.id           39.4132  2  2.764e-09 ***
## obs.year          3.0430  1    0.08109 .
## hydrology        29.9967  2  3.064e-07 ***
## esy4:site.type    9.0403  4    0.06010 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
summary(m_2)
```

```
## Linear mixed model fit by maximum likelihood  ['lmerMod']
## Formula: y ~ esy4 * site.type + eco.id + obs.year + hydrology + (1 | id.site)
##    Data: sites
##
##      AIC      BIC   logLik -2*log(L)  df.resid
##     5538     5609    -2753      5506       605
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -3.1782 -0.5504 -0.0016  0.5039  3.3962
##
## Random effects:
##  Groups   Name        Variance Std.Dev.
##  id.site  (Intercept) 418.7    20.46
##  Residual             239.0    15.46
## Number of obs: 621, groups:  id.site, 177
##
## Fixed effects:
##                      Estimate Std. Error t value
## (Intercept)          211.7360     4.8535  43.625
## esy4R22                2.9293     2.5382   1.154
## esy4R1A              -13.3948     5.1132  -2.620
## site.type.L           20.8033     4.3645   4.766
## site.type.Q            6.1834     3.1590   1.957
```

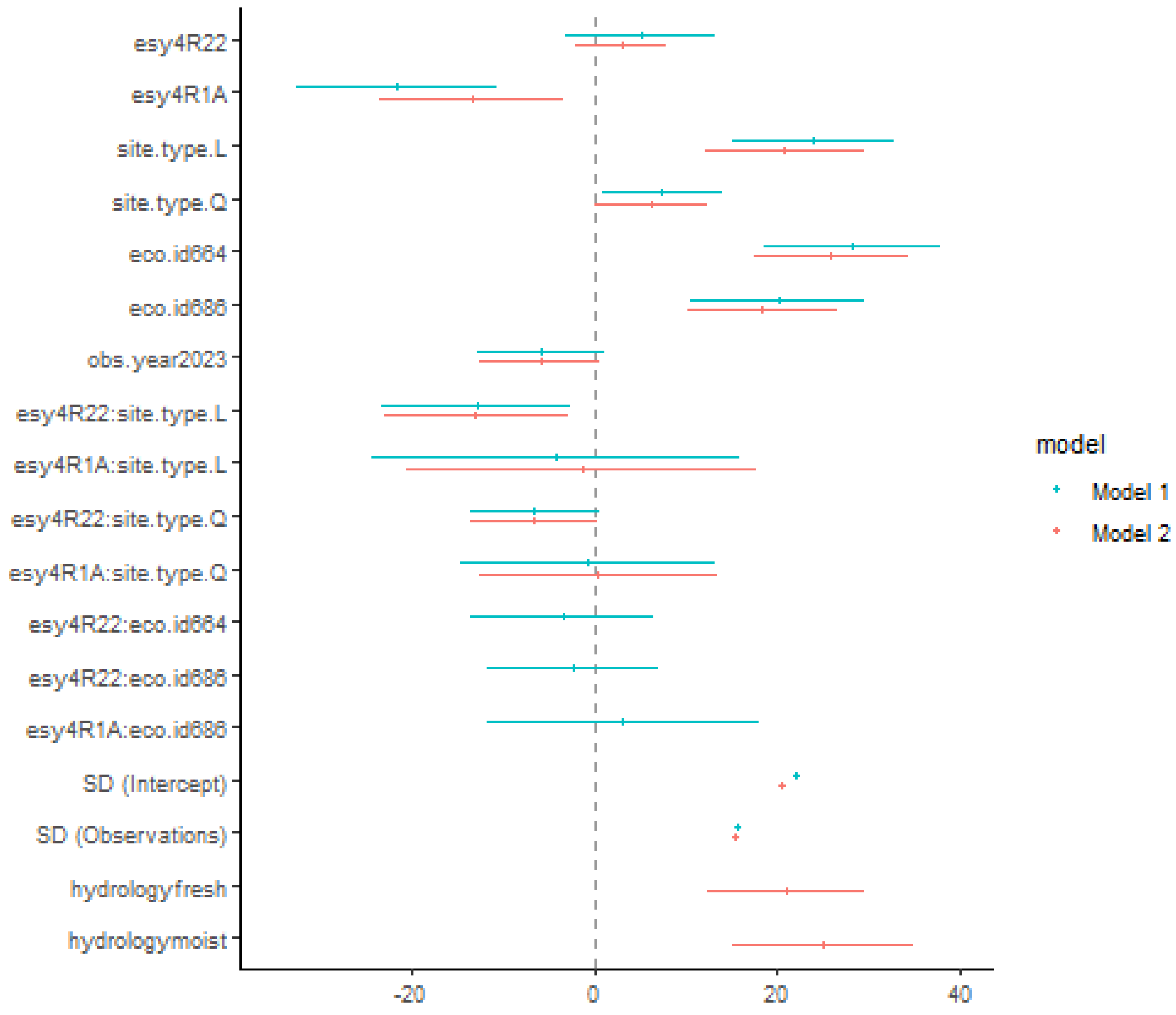
```
## eco.id664            25.8684     4.2421   6.098
## eco.id686            18.3211     4.1408   4.425
## obs.year2023         -5.9023     3.3835  -1.744
## hydrologyfresh       20.9107     4.3766   4.778
## hydrologymoist       24.9844     5.0251   4.972
## esy4R22:site.type.L -12.9476     5.1252  -2.526
## esy4R1A:site.type.L  -1.3590     9.7365  -0.140
## esy4R22:site.type.Q  -6.5424     3.4948  -1.872
## esy4R1A:site.type.Q   0.4328     6.6125   0.065
```

```
##
## Correlation matrix not shown by default, as p = 14 > 12.
## Use print(x, correlation=TRUE)  or
##     vcov(x)        if you need it
```

Forest plot

```
dotwhisker::dwplot(
  list(m_1, m_2),
  ci = 0.95,
  show_intercept = FALSE,
  vline = geom_vline(xintercept = 0, colour = "grey60", linetype = 2)) +
  xlim(-35, 40) +
  theme_classic()
```

```
## Package 'merDeriv' needs to be installed to compute confidence intervals
##   for random effect parameters.
## Package 'merDeriv' needs to be installed to compute confidence intervals
##   for random effect parameters.
```



Effect sizes

Effect sizes of chosen model just to get exact values of means etc. if necessary.

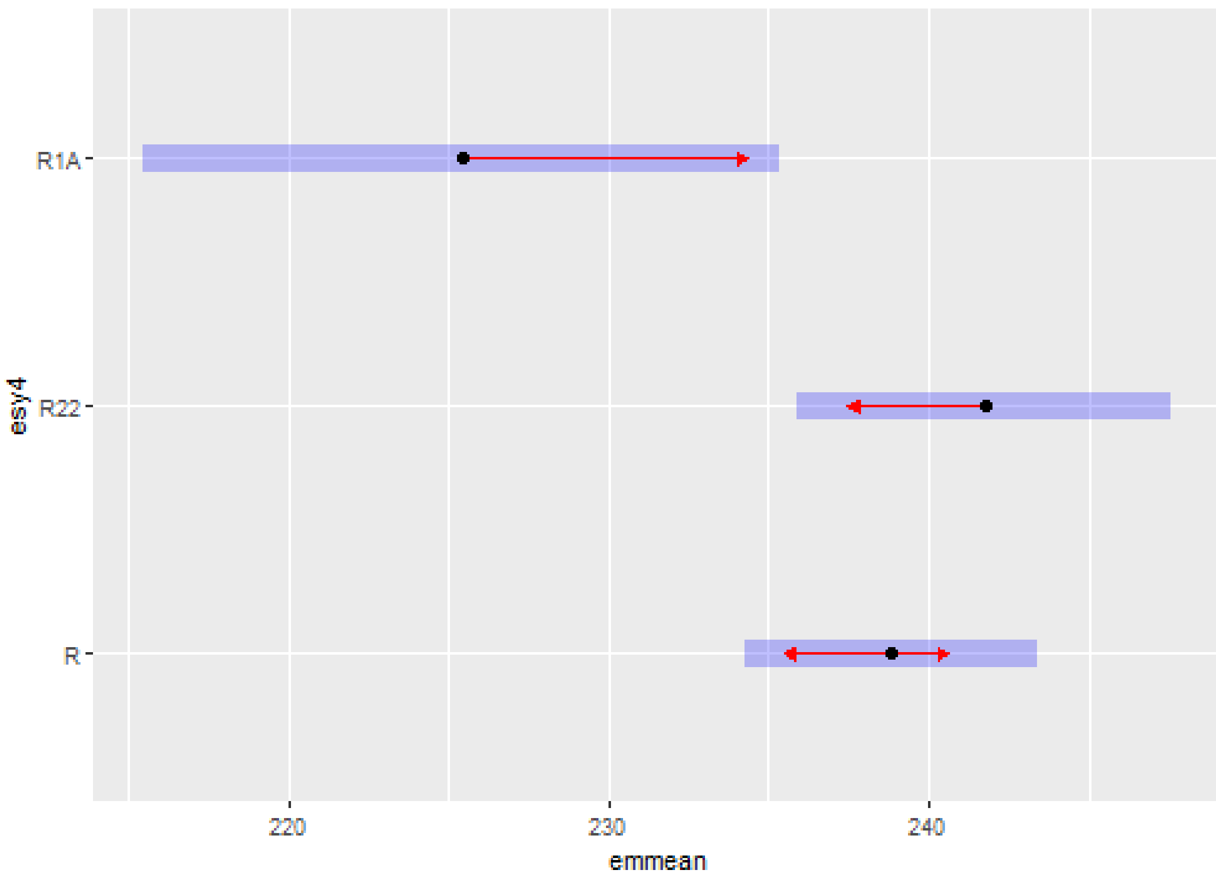
```
(emm <- emmeans(
  m_2,
  revpairwise ~ esy4,
  type = "response"
  ))
```

ESY EUNIS Habitat type

```
## $emmeans
##  esy4 emmean   SE  df lower.CL upper.CL
##  R       239 2.31 155      234      243
##  R22     242 2.97 282      236      248
##  R1A     225 5.04 316      216      235
##
## Results are averaged over the levels of: site.type, eco.id, obs.year, hydrology
```

```
## Degrees-of-freedom method: kenward-roger
## Confidence level used: 0.95
##
## $contrasts
##  contrast  estimate   SE  df t.ratio p.value
##  R22 - R       2.93 2.56 556   1.143  0.4880
##  R1A - R     -13.39 5.20 306  -2.574  0.0283
##  R1A - R22   -16.32 5.62 344  -2.906  0.0108
##
## Results are averaged over the levels of: site.type, eco.id, obs.year, hydrology
## Degrees-of-freedom method: kenward-roger
## P value adjustment: tukey method for comparing a family of 3 estimates
```

```r
plot(emm, comparison = TRUE)
```



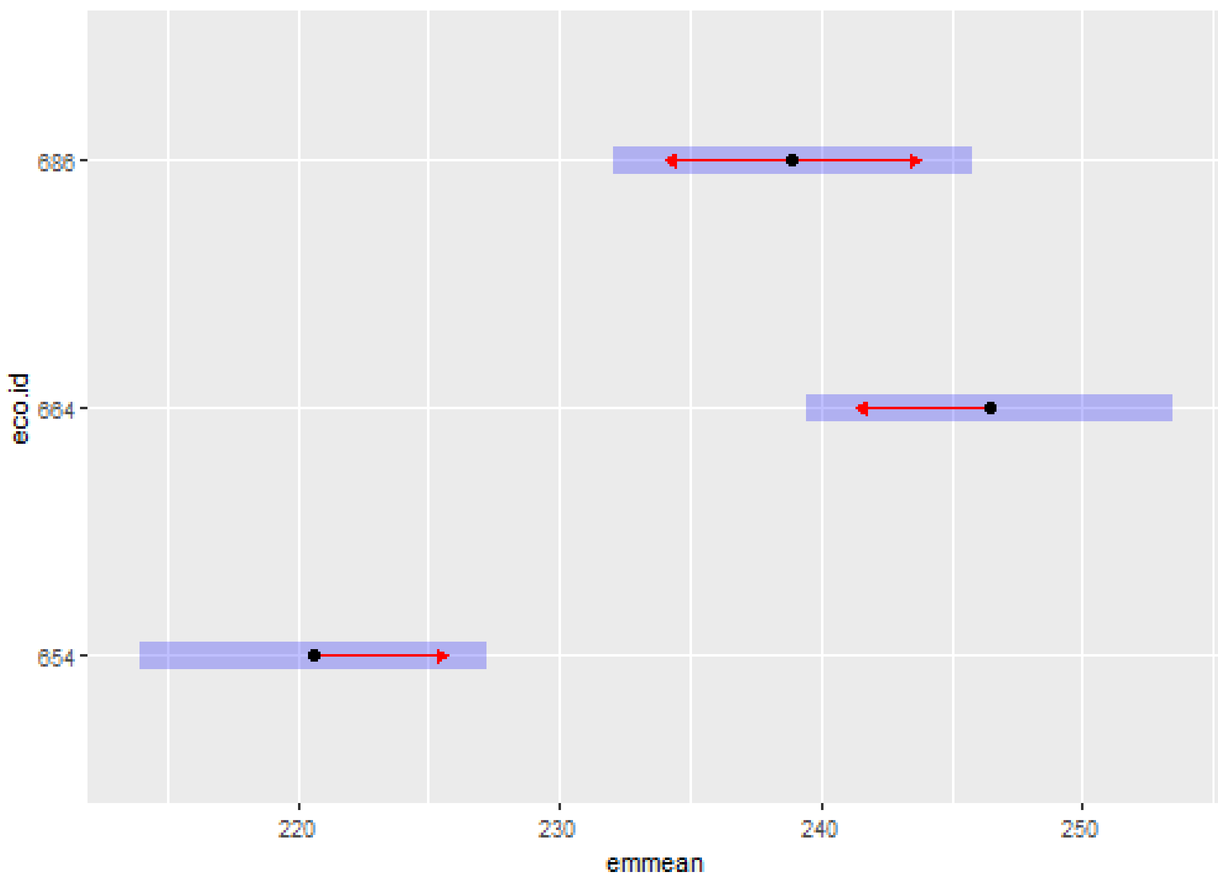

```r
(emm <- emmeans(
  m_2,
  revpairwise ~ eco.id,
  type = "response"
  ))
```

Habiat type x Region

```
## $emmeans
##  eco.id emmean   SE  df lower.CL upper.CL
```

```
##   654       221 3.37 153      214      227
##   664       246 3.56 226      239      253
##   686       239 3.50 186      232      246
##
## Results are averaged over the levels of: esy4, site.type, obs.year, hydrology
## Degrees-of-freedom method: kenward-roger
## Confidence level used: 0.95
##
## $contrasts
##   contrast              estimate   SE  df t.ratio p.value
##   eco.id664 - eco.id654    25.87 4.36 186   5.933  <.0001
##   eco.id686 - eco.id654    18.32 4.26 181   4.302  0.0001
##   eco.id686 - eco.id664    -7.55 4.24 185  -1.779  0.1794
##
## Results are averaged over the levels of: esy4, site.type, obs.year, hydrology
## Degrees-of-freedom method: kenward-roger
## P value adjustment: tukey method for comparing a family of 3 estimates
```

```r
plot(emm, comparison = TRUE)
```



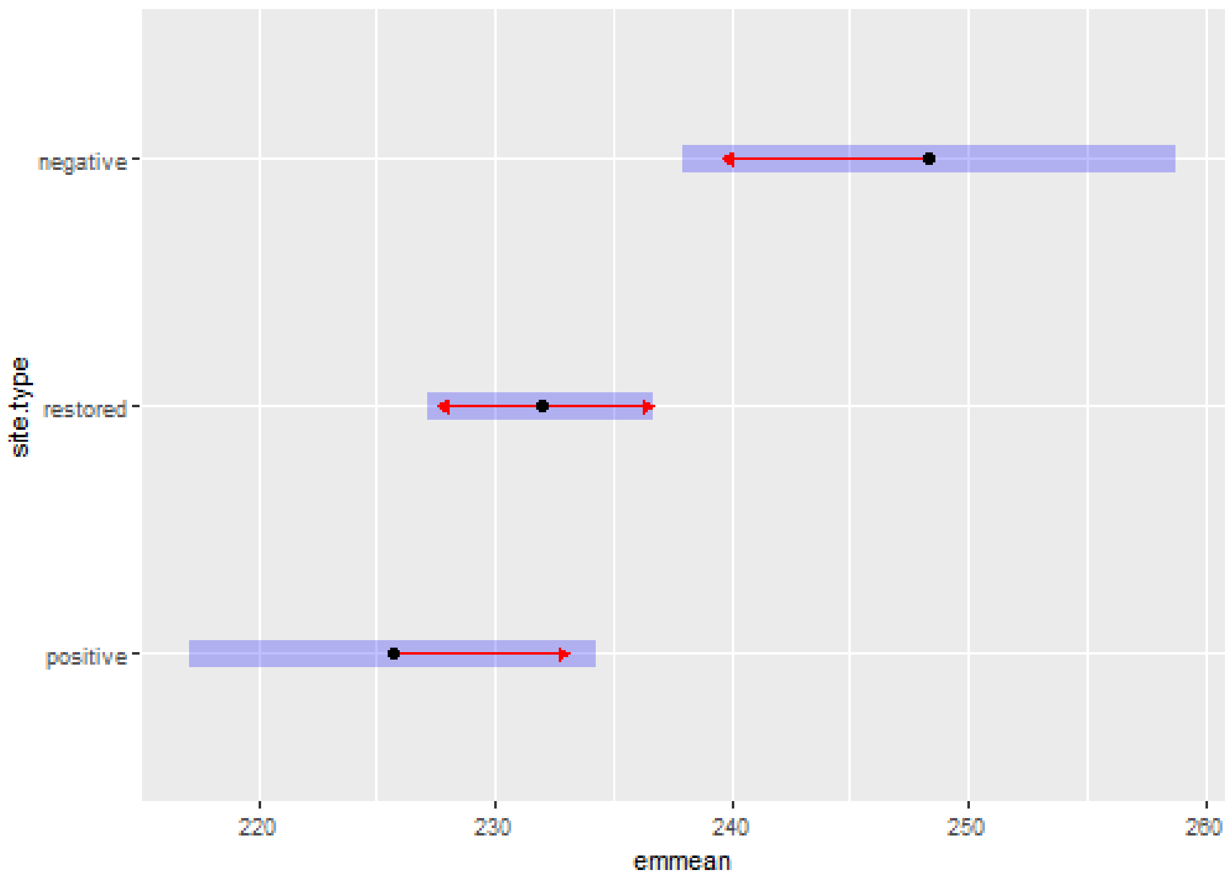

```r
(emm <- emmeans(
  m_2,
  revpairwise ~ site.type,
  type = "response"
```

```
  ))
```

Habiat type x Site type

```
## $emmeans
##  site.type emmean   SE  df lower.CL upper.CL
##  positive     226 4.37 204      217      234
##  restored     232 2.41 128      227      237
##  negative     248 5.30 241      238      259
##
## Results are averaged over the levels of: esy4, eco.id, obs.year, hydrology
## Degrees-of-freedom method: kenward-roger
## Confidence level used: 0.95
##
## $contrasts
##  contrast            estimate   SE  df t.ratio p.value
##  restored - positive     6.26 4.99 184   1.254  0.4233
##  negative - positive    22.68 6.90 221   3.288  0.0034
##  negative - restored    16.42 5.74 266   2.859  0.0127
##
## Results are averaged over the levels of: esy4, eco.id, obs.year, hydrology
## Degrees-of-freedom method: kenward-roger
## P value adjustment: tukey method for comparing a family of 3 estimates
```

```
plot(emm, comparison = TRUE)
```



Session info

```
## R version 4.5.0 (2025-04-11 ucrt)
## Platform: x86_64-w64-mingw32/x64
## Running under: Windows 11 x64 (build 26100)
##
## Matrix products: default
##   LAPACK version 3.12.1
##
## locale:
## [1] LC_COLLATE=German_Germany.utf8  LC_CTYPE=German_Germany.utf8
## [3] LC_MONETARY=German_Germany.utf8 LC_NUMERIC=C
## [5] LC_TIME=German_Germany.utf8
##
## time zone: America/New_York
## tzcode source: internal
##
## attached base packages:
## [1] stats     graphics  grDevices utils     datasets  methods   base
##
## other attached packages:
##  [1] emmeans_1.11.1   DHARMa_0.4.7     patchwork_1.3.0  ggbeeswarm_0.7.2
##  [5] lubridate_1.9.4  forcats_1.0.0    stringr_1.5.1    dplyr_1.1.4
##  [9] purrr_1.0.4      readr_2.1.5      tidyr_1.3.1      tibble_3.2.1
## [13] ggplot2_3.5.2    tidyverse_2.0.0  here_1.0.1
##
## loaded via a namespace (and not attached):
##  [1] Rdpack_2.6.4          gridExtra_2.3         sandwich_3.1-1
##  [4] rlang_1.1.6           magrittr_2.0.3        multcomp_1.4-28
##  [7] compiler_4.5.0        mgcv_1.9-1            vctrs_0.6.5
## [10] pkgconfig_2.0.3       crayon_1.5.3          fastmap_1.2.0
## [13] backports_1.5.0       labeling_0.4.3        utf8_1.2.5
## [16] ggstance_0.3.7        promises_1.3.2        rmarkdown_2.29
## [19] tzdb_0.5.0            nloptr_2.2.1          bit_4.6.0
## [22] xfun_0.52             later_1.4.2           broom_1.0.8
## [25] parallel_4.5.0        R6_2.6.1              gap.datasets_0.0.6
## [28] stringi_1.8.7         qgam_2.0.0            RColorBrewer_1.1-3
## [31] car_3.1-3             boot_1.3-31           estimability_1.5.1
## [34] Rcpp_1.0.14           iterators_1.0.14      knitr_1.50
## [37] zoo_1.8-14            parameters_0.25.0     httpuv_1.6.16
## [40] Matrix_1.7-3          splines_4.5.0         timechange_0.3.0
## [43] tidyselect_1.2.1      rstudioapi_0.17.1     abind_1.4-8
## [46] yaml_2.3.10           MuMIn_1.48.11         doParallel_1.0.17
## [49] codetools_0.2-20      lattice_0.22-6        plyr_1.8.9
## [52] bayestestR_0.15.3     shiny_1.10.0          withr_3.0.2
## [55] evaluate_1.0.3        marginaleffects_0.25.1 survival_3.8-3
## [58] pillar_1.10.2         gap_1.6               carData_3.0-5
## [61] foreach_1.5.2         stats4_4.5.0          reformulas_0.4.1
## [64] insight_1.2.0         generics_0.1.4        vroom_1.6.5
## [67] rprojroot_2.0.4       hms_1.1.3             scales_1.4.0
## [70] minqa_1.2.8           xtable_1.8-4          glue_1.8.0
## [73] tools_4.5.0           data.table_1.17.2     lme4_1.1-37
## [76] mvtnorm_1.3-3         grid_4.5.0            rbibutils_2.3
## [79] datawizard_1.1.0      nlme_3.1-168          Rmisc_1.5.1
```

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## [82] performance_0.13.0     beeswarm_0.4.0         vipor_0.4.7
## [85] Formula_1.2-5          cli_3.6.5              gtable_0.3.6
## [88] digest_0.6.37          pbkrtest_0.5.4         TH.data_1.1-3
## [91] farver_2.1.2           htmltools_0.5.8.1      lifecycle_1.0.4
## [94] mime_0.13              bit64_4.6.0-1          dotwhisker_0.8.4
## [97] MASS_7.3-65
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